

CLI Password Manager using AES-256 Encryption

Project Overview

The CLI Password Manager is a secure command-line application developed in Python that allows users to store, retrieve, update, and delete credentials securely.

The application uses AES-256-GCM encryption to protect sensitive information and PBKDF2-HMAC for secure key derivation from a master password.

All credentials remain encrypted at rest and are stored in a JSON-based vault.

Features

Secure Authentication

- Master Password Protection
- PBKDF2-HMAC Key Derivation
- Brute Force Protection

Credential Management

- Add New Credentials
- Retrieve Credentials
- List Stored Websites
- Delete Credentials

Encryption

- AES-256-GCM Encryption
- Authentication Tags
- Nonce Generation

Secure Storage

- JSON-Based Vault
 - Atomic File Writes
 - Data Integrity Validation
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Technologies Used

Technology	Purpose
Python	Programming Language
Cryptography	AES-256 Encryption

Technology	Purpose
Argparse	CLI Interface
JSON	Data Storage
Getpass	Hidden Password Input

Project Structure

```
CLI-Password-Manager-AES256/  
|  
├─ main.py  
├─ vault.py  
├─ crypto_utils.py  
├─ auth.py  
├─ vault.json  
├─ requirements.txt  
├─ screenshots/  
├─ README.md  
└─ Internship_Project_Report.pdf
```

Installation

Clone Repository

```
git clone https://github.com/Umahika/CLI-Password-Manager-AES256.git
```

Install Dependencies

```
pip install -r requirements.txt
```

Running the Application

Add Credentials

```
python main.py add
```

Retrieve Credentials

```
python main.py get
```

List Stored Sites

```
python main.py list
```

Delete Credentials

```
python main.py delete
```

Example Output

```
Enter Master Password:
```

```
Website: github.com
```

```
Username: user123
```

```
Password: *****
```

```
Credential Added Successfully
```

Security Features

- AES-256-GCM Encryption
- PBKDF2-HMAC Key Derivation
- Random Salt Generation
- Nonce-Based Encryption
- Authentication Tag Validation
- Secure Password Input

Testing Results

Test Case	Status
Master Password Verification	Passed
Credential Encryption	Passed
Credential Decryption	Passed
Add Operation	Passed

Test Case	Status
Delete Operation	Passed
JSON Storage	Passed

Future Enhancements

- Password Generator
 - Multi-User Support
 - Cloud Backup
 - GUI Version
 - Password Strength Analysis
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Learning Outcomes

- Cryptographic Security Principles
 - AES-256 Encryption
 - CLI Application Development
 - JSON Data Management
 - Secure Authentication Systems
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Author

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License

This project is developed for educational and internship purposes only.